3.2 Implicit Differentiation

You missed this lesson. Copy this into your notes. That is all you need to do.

Question of the day

How do you find the slope of a curve that is not a function?

Targets

- I can differentiate both sides of an equation with x and y.
- I can solve for dy/dx and find a slope at a point.
- I can write the tangent line at a point on the curve.

Concepts

- **Implicit form** ($x^2 + y^2 = 25$): x and y are mixed in one equation. Near a point, y still behaves as a function of x.
- **The y-factor** ($d/dx\ (y^2) = 2y \cdot dy/dx$): y depends on x, so every y-term is a composite. Differentiate it, then multiply by dy/dx.
- **Products** ($d/dx\ (xy) = y + x \cdot dy/dx$): A term with x and y needs the product rule. Both parts give a piece.
- **Collect, factor, divide** ($dy/dx = -(2x) / (2y)$): Move every dy/dx term to one side. Factor out dy/dx, then divide.
- **A slope needs a point** ($y - b = (dy/dx)(x - a)$): The answer has x and y in it. Use both coordinates of the point on the curve.

What the class predicted, and what is true

- The circle $x^2 + y^2 = 25$ passes through $(3, 4)$ and $(3, -4)$. The tangent slopes at those two points are... **opposite in sign.** The upper arc falls at $x = 3$ and the lower arc rises there: the slopes are $-\frac{3}{4}$ and $+\frac{3}{4}$. One x cannot name the slope.
- Suppose $y = x^3$. Then $y^2 = x^6$, so $\frac{d}{dx}(y^2) = 6x^5$. Which expression gives $6x^5$? $2y \cdot \frac{dy}{dx}$. $2y \cdot \frac{dy}{dx} = 2x^3 \cdot 3x^2 = 6x^5$. The chain rule puts $\frac{dy}{dx}$ on every y-term.

Huddle notes (answers filled in)

Definitions

- **Implicit form:** An equation where x and y are **mixed** together.
- **The y-factor:** Every y-term gets a factor of **dy/dx**.
- **Solving:** Collect the dy/dx terms, **factor** it out, then divide.

Worked example

Find dy/dx for $x^2 + y^2 = 25$.

- Differentiate both sides: $2x + 2y \cdot dy/dx =$ **0**
- Isolate: $2y \cdot dy/dx =$ **$-2x$**
- Divide by 2y: $dy/dx =$ **$-x/y$**

Label the tangent

1. The point (3, 4) lies on the **curve**.
2. The **tangent line** touches the curve at that point.
3. Its slope at (3, 4) is **$-3/4$**.

One worked sample of each skill

Find dy/dx

Level 1. Differentiate both sides, then solve for $\frac{dy}{dx}$.
Answer: $\frac{4}{2y}$ $2y\,\dfrac{dy}{dx} = 4$, so $\dfrac{dy}{dx} = \dfrac{4}{2y}$.

Level 2. Differentiate both sides, then solve for $\frac{dy}{dx}$.
Answer: $\frac{-x}{2y}$ $2x + 4y\,\dfrac{dy}{dx} = 0$, so $\dfrac{dy}{dx} = \dfrac{-2x}{4y} = -\dfrac{x}{2y}$.

Level 3. Differentiate both sides. Collect the $\frac{dy}{dx}$ terms, factor, and divide.
Answer: $\frac{5x^4-4x}{9y^2-1}$ $9y^2y' + 4x - y' = 5x^4$, so $y'(9y^2 - 1) = 5x^4 - 4x$.

Level 4. Differentiate both sides. The right side needs the product rule.
Answer: $\frac{6y-3x^2}{3y^2-6x}$ $3x^2 + 3y^2y' = 6y + 6x\,y'$, so $y'(3y^2 - 6x) = 6y - 3x^2$.

Differentiate a y-term

Level 1. $\dfrac{d}{dx}\left(y^3\right)$ equals...
(A) $3y^2 \cdot dy/dx$ (B) $3x^2$ (C) $3y^2 \cdot dx/dy$ (D) $3y^2$
Answer: A Power rule gives $3y^2$. The chain rule adds the factor $\frac{dy}{dx}$.

Level 2. $\dfrac{d}{dx}\left(6y^2\right)$ equals...
(A) 12y (B) $6y \cdot dy/dx$ **(C) $12y \cdot dy/dx$** (D) 12x
Answer: C Coefficient: 6 times 2 is 12. Then the factor $\frac{dy}{dx}$.

Level 3. $\dfrac{d}{dx}\left(x^3y\right)$ equals...
(A) $3x^2y$ (B) $3x^2 \cdot dy/dx$ (C) $x^3 \cdot dy/dx$ **(D) $3x^2y + x^3 \cdot dy/dx$**
Answer: D Product rule: (derivative of the first)(second) + (first)(derivative of the second). The y part brings $\frac{dy}{dx}$.

Level 4. $\dfrac{d}{dx}\left(x^2e^y\right)$ equals...
(A) x^2 · e^y · dy/dx (B) 2x · e^y **(C) 2x · e^y + x^2 · e^y · dy/dx** (D) 2x · e^y · dy/dx
Answer: C Product rule: $2x\,e^y$ plus $x^2e^y\frac{dy}{dx}$.

Slope at a point

Level 1. The point $(6,-8)$ is on the curve $x^2+y^2=100$. Find $\dfrac{dy}{dx}$ there.
Answer: 0.75 Implicit differentiation gives a slope formula in x and y. At $(6,-8)$ it equals $\dfrac{3}{4}$.

Level 2. The point $(-1,2)$ is on the curve $x^2+2y^2=9$. Find $\dfrac{dy}{dx}$ there.
Answer: 0.25 Implicit differentiation gives a slope formula in x and y. At $(-1,2)$ it equals $\dfrac{1}{4}$.

Level 3. The point $(4,-1)$ is on the curve $x^2+xy+y^2=13$. Find $\dfrac{dy}{dx}$ there.
Answer: -3.5 Implicit differentiation gives a slope formula in x and y. At $(4,-1)$ it equals $\dfrac{-7}{2}$.

Level 4. The point $(1,4)$ is on the curve $6xy=8+y^2$. Find $\dfrac{dy}{dx}$ there.
Answer: 12 Implicit differentiation gives a slope formula in x and y. At $(1,4)$ it equals 12.

Tangent line

Level 1. Write the tangent line to the curve at the point.
Answer: $\frac{5}{12}\left(x-(-5)\right)+12$ Slope at the point: $\dfrac{5}{12}$. Then $y=m(x-(-5))+(12)$.

Level 2. Write the tangent line to the curve at the point.
Answer: $\frac{1}{4}\left(x-(-1)\right)+2$ Slope at the point: $\dfrac{1}{4}$. Then $y=m(x-(-1))+(2)$.

Level 3. Write the tangent line to the curve at the point.
Answer: $\frac{-7}{2}\left(x-4\right)-1$ Slope at the point: $\dfrac{-7}{2}$. Then $y=m(x-(4))+(-1)$.

Level 4. Write the tangent line to the curve at the point. Keep the radicals exact.
Answer: $\frac{3}{\sqrt{7}}\left(x-3\right)-\sqrt{7}$ Slope: $-\dfrac{3}{-\sqrt{7}}$. Then $y=m(x-3)+(-\sqrt{7})$.